

L1 — YAMNet Baseline Scores

plan.md leg: "Yamnet Baseline Scores". **Goal:** see what raw (un-fine-tuned) YAMNet already thinks about our sounds. **Script:** `experiments/scripts/leg1_baseline.py` (reads the cache from `extract_yamnet.py`). **Artifacts:** `experiments/outputs/baseline_topk.csv`, `baseline_summary.json`, `figures/baseline_top1_hist.png`.

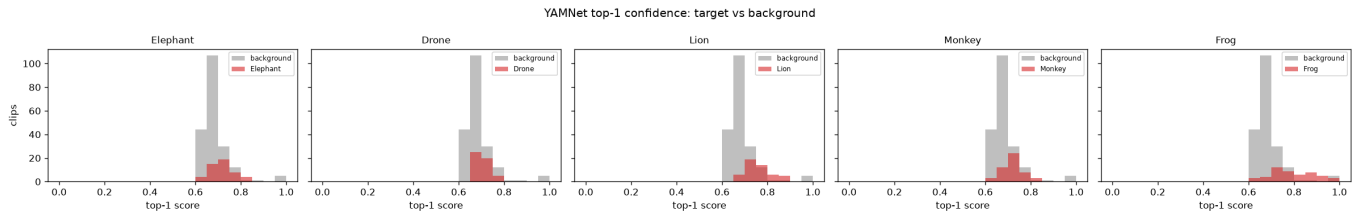
Method

Ran the project's **custom mel-patch YAMNet** (`yamnet/export_out/tf2`, 521 AudioSet classes) on all 450 clips. This is the same model ODAS calls on-device — it takes 96×64 log-mel patches derived from the 257-bin spectra (front-end: STFT 400/160/512 → 64 mel 125–7500 Hz → log → 96×48 patches, matching `data_loader.py` and the C wrapper). For each clip we mean-pool the per-patch scores and take the top-5 AudioSet classes.

Finding: raw YAMNet does **not** recognize our classes

class	most common top-1	sensible?	mean top-1 score	% clips w/ relevant class in top-5
Elephant	Jackhammer (8/50)	✗	0.72	12%
Drone	Jackhammer (13/50)	✗	0.70	0%
Lion	Cupboard open/close (16/50)	✗	0.76	2%
Monkey	Rail transport (4/50)	✗	0.72	2%
Frog	Eruption (8/50)	✗	0.80	0%

The top-1 predictions are **acoustically adjacent but wrong**: elephant rumble → *Jackhammer*, drone motor → *Jackhammer*, frog chorus → *Eruption*. Almost nothing lands on an animal/relevant AudioSet class.



The histogram shows target and background clips both cluster at mid-high top-1 confidence (0.6–0.8) on these wrong classes — so **raw top-1 confidence does not separate target from background**. The 521-class head is uncalibrated for this domain.

Interpretation (per plan.md's key insight)

"If YAMNet already scores elephant clips high on Animal/Elephant — your job is easier. If it scores them high on Music — the embedding space needs work."

Here the *head* is useless out-of-the-box, but the next leg (L2/ L3) shows the *embedding space* is excellent. So the job is **easy**: a lightweight probe / fine-tuned head recovers the classes — no backbone surgery needed. This is the direct empirical motivation for the fine-tuning program in [experiments.pdf](#) (Phase 2).

Pass criterion ("you understand what YAMNet already knows"):  — it knows essentially nothing at the label level for these classes, which is itself the actionable result.